



ParulTM
University



Parul University

**FACULTY OF ENGINEERING AND
TECHNOLOGY OF TECHNOLOGY**

High
Performance
Computing
(303105356)

7th SEMESTER

Laboratory Manual

Enrollment No: 2203031050492



ParulTM
University

CERTIFICATE

This is to certify that **CHILAKABATHULA KESHAVARDHAN** with
Enrolment no. **2203031050492** an **7th Semester CSE9** has successfully
completed her laboratory experiments in the **High Performance Computing (303105356)** from the
department of **Computer Science & Engineering** during the academic year **2025-26**



Date of Submission:

Staff In charge:

Head Of Department:

Enrollment No: 2203031050492

TABLE OF CONTENT

Sr. No	Experiment Title	Page No		Date of Start	Date of Completion	Sign	Marks (out of 10)
		From	To				
1	Study the facilities provided by Google Colab						
2	Demonstrate basic Linux Commands.						
3	Using Divide and Conquer Strategies design a class for Concurrent Quick Sort using C++.						
4	Demonstrate MPI functions through a simple program.						
5	Write a program to check task distribution using Gprof.						
6	Write a simple CUDA program to print "Hello World!"						
7	Write a simple CUDA program to add two numbers.						
8	Write a CUDA program to add two arrays.						
9	Analyze the code using Nvidia-Profilers.						
10	Demonstration of OpenMP and pthread functions.						

PRACTICAL-01

AIM: Study the facilities provided by Google Colab.

1) Exclusive Features of Google Colab:

1. Free Access to GPUs and TPUs

- You can run your code on free GPUs (like NVIDIA Tesla K80, T4) or TPUs, which speeds up heavy computations such as deep learning training — no setup required.

2. Cloud-Based Jupyter Notebooks

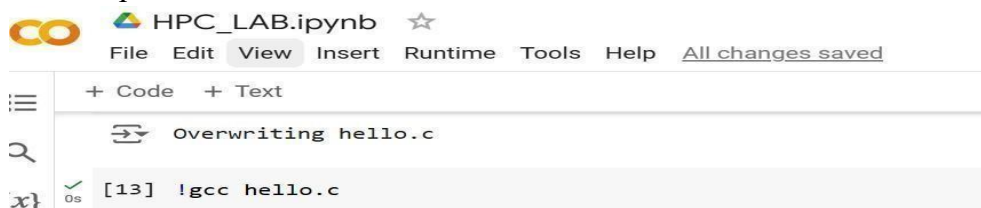
- Your notebooks run entirely in the cloud — no need to install anything locally.
- You can share notebooks easily via Google Drive or links.

3. Seamless Integration with Google Drive

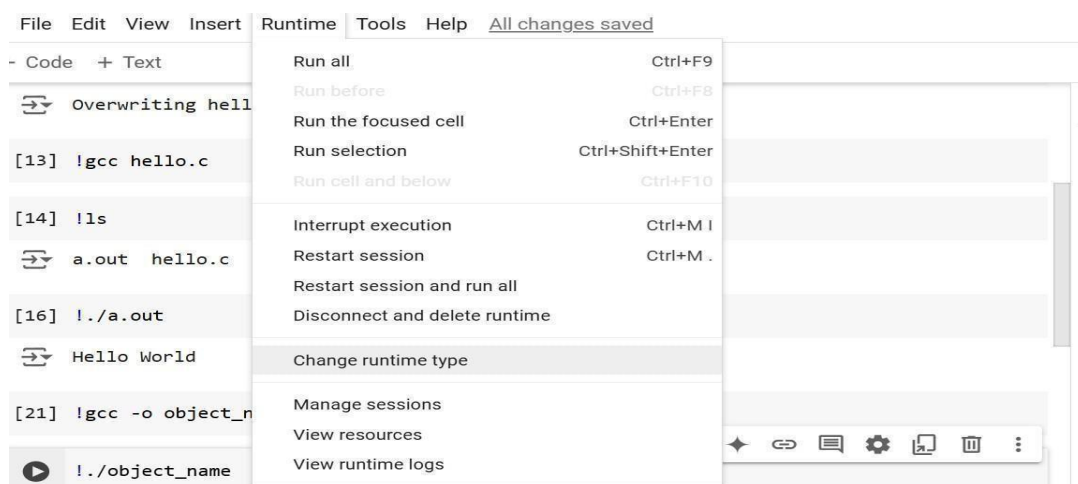
- Save and load notebooks directly from your Google Drive.
- Mount your Drive to read/write files within the Colab environment.

2) How to Change runtime type in google colab:

- Open Runtime menu.



- Navigate to Change runtime type and click.



- Choose whether you need CPU/GPU needed for execution.

Change runtime type

Runtime type

Python 3

Hardware accelerator ?

☒ CPU ☐ T4 GPU ☐ A100 GPU ☐ L4 GPU

☐ v2-8 TPU

Want access to premium GPUs? [Purchase additional compute units](#)

Cancel Save

3) PYTHON program execution in Google Colab.

- Program to find no of vowels

PRACTICAL-02 - Colab x Colab Notebooks - Google Drive x Hello world.ipynb - Colab x +

colab.research.google.com/drive/15UD1FG0G-L9n4cV8xB5qfoR5qqxVbhlh#scrollTo=6jOPEWDil2WC

Gmail YouTube Maps All Bookmarks

Hello world.ipynb ☆

File Edit View Insert Runtime Tools Help

Q Commands + Code + Text ▶ Run all ✓ RAM Disk

```
[1] def count_vowels(text):
    vowels = "aeiou"
    count = 0
    for char in text.lower():
        if char in vowels:
            count += 1
    return count

s = input("Enter a string: ")
print("Number of vowels:", count_vowels(s))
```

Enter a string: KESHAVARDHAN
Number of vowels: 4



4.C Program in Google Colab.

Sum of Numbers

The screenshot shows a Google Colab notebook titled 'Hello world.ipynb'. The code editor contains a C program to calculate the sum of two integers. The program includes `<stdio.h>`, defines `main()` with variables `a`, `b`, and `sum`, prompts the user for two integers, reads them using `scanf`, calculates the sum, and prints it using `printf`. The program is saved as `add.c`. Below the code editor, a terminal window shows the execution of `gcc add.c -o add` and `echo "4 7" | ./add`, resulting in the output 'Sum = 11'.

```
%writefile add.c
#include <stdio.h>

int main() {
    int a, b, sum;

    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);

    sum = a + b;
    printf("Sum = %d\n", sum);

    return 0;
}
```

Writing add.c

```
!gcc add.c -o add
!echo "4 7" | ./add
```

Enter two integers: Sum = 11

Conclusion:

Using Google Colab lets run and test C or Python programs quickly without setup hassles. Its free cloud GPUs and easy sharing boost learning and collaboration. Overall, it's a powerful, accessible platform for coding and experimentation.